TryHackMe - Nmap Live Host Discovery

Platform: TryHackMe
Room link: Nmap Live Host Discovery

1. Introduction

Live Host Discovery is an important step in network reconnaissance. Before scanning ports or identifying services, we need to determine which devices are active on a network.
Nmap provides different techniques to discover live hosts using ARP, ICMP, TCP, and UDP packets.

In this room, we explore how Nmap identifies active systems and how different network protocols can be used to determine whether a target is online.

The room also introduces two additional network scanning tools: arp-scan and masscan.

2. Objectives

The main objectives of this room are:

- Understand subnetworks and IP addressing.
- Learn how Nmap handles target enumeration.
- Understand ARP-based host discovery.
- Perform ICMP-based host discovery.
- Understand TCP and UDP ping scans.
- Learn about reverse DNS lookups.
- Understand the functionality of arp-scan.
- Explore Masscan as a high-speed network scanner.
- Analyse network traffic using Wireshark.

3. Tools and Requirements

Tool	Purpose
Nmap	Network discovery and scanning
arp-scan	Discover devices using ARP
Masscan	High-speed port scanning
Wireshark	Network packet analysis
Kali Linux	Penetration testing environment
TryHackMe AttackBox	Online testing environment

4. Task-wise Walkthrough

Task 1 - Introduction

Nmap is an open-source network scanning tool used to discover devices and services across a network.
During reconnaissance, two important questions need to be answered:

- Which systems are currently online?
- Which services are running on those systems?

The first question is addressed through live host discovery, while the second is addressed through port scanning and service enumeration.

Nmap performs several operations depending on the options supplied.

A typical scanning workflow can include:

- Target enumeration.
- Host discovery.
- Reverse DNS lookup.
- Port scanning.
- Service version detection.
- Operating system detection.
- Traceroute.
- NSE script execution.
- Saving scan results.

Understanding this workflow helps us perform network reconnaissance systematically.

Task 2 - Subnetworks

A subnetwork, commonly called a subnet, is a logically divided portion of an IP network.

Subnetting allows network administrators to divide a large network into smaller networks.

An IPv4 address consists of 32 bits and is generally represented in four octets.
Example:192.168.1.10

A subnet mask determines which portion of the IP address represents the network and which portion represents the host.

A common example is: 192.168.1.0/24
Here, /24 represents the network prefix length.

The subnet mask is: 255.255.255.0

Important concepts:

- IP address
- Subnet mask
- Network address

- Broadcast address
- Host address
- CIDR notation

Subnetting is important in network scanning because it helps determine the range of IP addresses that need to be investigated.

Task 3 - Enumerating Targets

Nmap supports multiple ways of specifying target systems.

i. Scanning a single IP address

```
nmap 192.168.1.10
```

ii. Scanning multiple IP addresses

```
nmap 192.168.1.10 192.168.1.20 192.168.1.30
```

iii. Scanning an IP range

```
nmap 192.168.1.10-50
```

iv. Scanning an entire subnet

```
nmap 192.168.1.0/24
```

v. Scanning targets from a file

First, create a target list:

```
nano targets.txt
```

Add the authorized target IP addresses, one per line.

Then execute:

```
nmap -iL targets.txt
```

vi. Excluding specific targets

```
nmap 192.168.1.0/24 --exclude 192.168.1.10
```

These methods allow us to organize and perform scans against individual systems, multiple hosts, or entire network ranges.

Task 4 - Discovering Live Hosts

Host discovery is the process of identifying devices that are active and reachable on a network.

Nmap uses different protocols to determine whether a target is online.

The primary techniques covered in this room are:

i. ARP-based discovery.
ii. ICMP-based discovery.
iii. TCP-based discovery.

iv. UDP-based discovery.

ARP - Address Resolution Protocol
ARP is used to identify the MAC address associated with an IPv4 address on a local network.

ICMP - Internet Control Message Protocol
ICMP is used for network diagnostics and communication of network errors.

TCP - Transmission Control Protocol
TCP is a connection-oriented transport protocol that uses flags such as SYN, ACK, and RST.

UDP - User Datagram Protocol
UDP is a connectionless transport protocol that can be used for lightweight communication and host discovery.
Understanding these protocols helps us select suitable host discovery methods.

Task 5 - Nmap Host Discovery Using ARP
ARP stands for Address Resolution Protocol.

It is used to map IPv4 addresses to MAC addresses within a local network.

When a device wants to communicate with another device on the same subnet, it may first send an ARP request to discover the destination's MAC address.

The general ARP communication process:

i. A device sends an ARP request.
ii. The request is broadcast across the local network.
iii. The device with the matching IP address responds.
iv. The sender learns the destination MAC address.

Performing an ARP scan with Nmap

```
sudo nmap -PR -sn 192.168.1.0/24
```

Explanation:

- sudo - Executes Nmap with elevated privileges.
- -PR - Uses ARP requests for host discovery.
- -sn - Disables port scanning.
- 192.168.1.0/24 - Specifies the target subnet.

Using arp-scan

Another tool used for ARP-based discovery is arp-scan.

To identify the network interface:

ip addr

To perform an ARP scan:

sudo arp-scan --localnet

To specify a network interface:

sudo arp-scan -I eth0 --localnet

ARP scanning is particularly useful when the scanning machine and target devices are connected to the same local network.

Task 6 - Nmap Host Discovery Using ICMP

ICMP stands for Internet Control Message Protocol.

It is commonly used for network troubleshooting and host discovery.

The traditional ping command uses ICMP Echo Request and Echo Reply messages.

i. ICMP Echo Scan

Nmap uses the -PE option to send ICMP Echo Requests.

sudo nmap -PE -sn 192.168.1.0/24

ii. ICMP Timestamp Scan

Nmap supports ICMP Timestamp Requests through the -PP option.

sudo nmap -PP -sn 192.168.1.0/24

iii. ICMP Address Mask Scan

Nmap supports ICMP Address Mask Requests through the -PM option.

sudo nmap -PM -sn 192.168.1.0/24

These techniques use different ICMP message types to identify active systems. However, ICMP traffic may be blocked by firewalls, so a lack of response does not necessarily mean that a device is offline.

Task 7 - Nmap Host Discovery Using TCP and UDP

When ICMP requests are blocked, TCP and UDP probes can provide alternative methods of identifying active systems.

i. TCP SYN Ping

TCP SYN Ping sends TCP packets with the SYN flag enabled.

Command:

sudo nmap -PS80,443 -sn 192.168.1.0/24

The target may respond with:

- SYN-ACK if the destination port is open.
- RST if the destination port is closed.

Both responses indicate that the target is reachable.

ii. TCP ACK Ping

TCP ACK Ping sends TCP packets with the ACK flag enabled.

Command:

```
sudo nmap -PA80,443 -sn 192.168.1.0/24
```

This technique can help identify hosts when traditional ICMP discovery is unavailable.

iii. UDP Ping

UDP Ping sends UDP packets to specified ports and analyses the responses.

Command:

```
sudo nmap -PU53,161,162 -sn 192.168.1.0/24
```

Depending on the destination port and target configuration, the response may include an ICMP error message or no response.

4. Introduction to Masscan

Masscan is a high-speed network scanner designed to scan large networks quickly.
Its syntax is similar to Nmap.

Examples:

```
sudo masscan 192.168.1.0/24 -p443
sudo masscan 192.168.1.0/24 -p80,443
sudo masscan 192.168.1.0/24 -p22-25
sudo masscan 192.168.1.0/24 --top-ports 100
```

Masscan can generate a large volume of network traffic. Its scanning rate should be configured carefully to avoid overwhelming networks.

Task 8 - Using Reverse DNS Lookup

DNS stands for Domain Name System.

It translates domain names into IP addresses.

Reverse DNS lookup performs the opposite operation by attempting to identify the hostname associated with an IP address.

For example:

IP Address → Hostname

Nmap supports reverse DNS lookup during host discovery.

Disable DNS resolution

```
nmap -n 192.168.1.0/24
```

Force reverse DNS resolution

```
nmap -R 192.168.1.0/24
```

Specify a DNS server

```
nmap --dns-servers 8.8.8.8 192.168.1.0/24
```

Reverse DNS can provide useful information about network infrastructure, device roles, and naming conventions.

However, not every IP address has a configured reverse DNS record.

5. Practical Exercises

The following exercises help reinforce the concepts covered in this room.

Exercise 1 - Network Discovery

Identify the network interface and determine the connected subnet.

```
ip addr
```

Exercise 2 - ARP Discovery

Perform an ARP scan against the local network.

```
sudo nmap -PR -sn TARGET_SUBNET
```

Exercise 3 - ICMP Discovery

Perform an ICMP Echo scan.

```
sudo nmap -PE -sn TARGET_SUBNET
```

Exercise 4 - TCP SYN Ping

Perform TCP SYN-based host discovery.

```
sudo nmap -PS22,80,443 -sn TARGET_SUBNET
```

Exercise 5 - TCP ACK Ping

Perform TCP ACK-based host discovery.

```
sudo nmap -PA22,80,443 -sn TARGET_SUBNET
```

Exercise 6 - UDP Discovery

Perform UDP-based host discovery.

```
sudo nmap -PU53,161,162 -sn TARGET_SUBNET
```

Exercise 7 - Reverse DNS

Perform reverse DNS lookup during host discovery.

nmap -R -sn TARGET_SUBNET

Exercise 8 - Packet Analysis

Open Wireshark and capture network traffic while performing the different discovery scans. Observe:

- ARP requests and replies.
- ICMP Echo Requests and Replies.
- TCP SYN and ACK packets.
- TCP RST responses.
- UDP packets.
- ICMP Destination Unreachable messages.

6. Important Commands Cheat Sheet

Command	Purpose
nmap -sn IP/24	Host discovery only
nmap -PR -sn IP/24	ARP discovery
nmap -PE -sn IP/24	ICMP Echo discovery
nmap -PP -sn IP/24	ICMP Timestamp discovery
nmap -PM -sn IP/24	ICMP Address Mask discovery
nmap -PS80,443 -sn IP/24	TCP SYN Ping
nmap -PA80,443 -sn IP/24	TCP ACK Ping
nmap -PU53 -sn IP/24	UDP Ping
nmap -n IP/24	Disable DNS resolution
nmap -R IP/24	Force reverse DNS lookup
nmap -iL targets.txt	Read targets from a file
arp-scan --localnet	Discover local network devices
masscan IP/24 -p80	High-speed port scanning

7. Key Takeaways

After completing this room, you should understand:

- How IP addresses and subnetworks work.
- How Nmap enumerates multiple targets.
- How ARP discovers devices on a local network.
- How ICMP is used for host discovery.
- How TCP SYN and ACK probes identify reachable systems.
- How UDP probes help discover active hosts.
- How Masscan performs high-speed network scanning.
- How reverse DNS lookup provides additional network information.
- How Wireshark can be used to analyse discovery traffic.

8. Conclusion

The Live Host Discovery room provides an understanding of how to identify active devices within a network using different network protocols.

It demonstrates that relying on a single discovery technique is not always effective because firewalls and network configurations can block certain types of traffic.

By learning ARP, ICMP, TCP, and UDP discovery techniques, we can perform more effective network reconnaissance and prepare for further enumeration.